Peter Mastnak

Curriculum Vitae

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RESEARCH INTERESTS

Scientific machine learning; physics-informed neural networks; probabilistic spatiotemporal modeling; approximate inference; causal modeling of environmental systems; data assimilation; climate and weather forecasting.

EDUCATION

PhD in Physical Oceanography 2024 – Present

Stanford University – Doerr School of Sustainability

Member of Dr. Barbara Block's lab at Hopkins Marine Station. Research sits at the intersection of climate science, marine ecology, and AI, with a focus on the concept of *animals as oceanographers* – using marine predators (tunas, sharks, seabirds, marine mammals) as autonomous sensing platforms that sample the ocean interior in ways fixed infrastructure cannot. Developing physics-informed neural networks and data assimilation frameworks that fuse animal-borne electronic tag data with Argo float profiles, ocean glider observations, CTD measurements, and satellite remote sensing to reconstruct subsurface ocean state and model climate-driven ecosystem dynamics. Building hierarchical behavioral classification pipelines and species distribution models that couple biological movement data with physical oceanographic fields. Applying GAM modeling and deep learning to longline fishing datasets for catch prediction under environmental variability. Research contributes to ocean digital twin development and informs international ocean policy and sustainable fisheries governance through evidence-based conservation frameworks.

MSc. Machine Learning and Data Science 2019 – 2021

KTH Royal Institute of Technology – Stockholm

Specialized in machine learning for environmental data analysis and predictive modeling. Coursework covered deep learning, probabilistic graphical models, Bayesian inference, computer vision, and scalable data pipelines. Thesis and project work focused on applying ML to geospatial and time-series environmental datasets. kth.se/machine-learning

BSc. Computer Science 2015 – 2018

University of Maribor – Slovenia

Focused on algorithmic development, data structures, and software engineering. Includes Erasmus exchange at University College Cork, Ireland (2017–2018) and LSE Management Summer School (2015).

WORK EXPERIENCE

BlueTraker – Maritime IoT & Sustainable Fisheries

https://www.bluetraker.com

ML Engineer & Product Lead Jan 2021 – Present

BlueTraker

Led end-to-end development of ocean-sensing products and ML systems across two concurrent roles. Developed species distribution and fishing activity classification models using VMS, AIS, and marine sensor data, integrating environmental covariates from ECMWF, ICON, and Azure-hosted oceanographic pipelines. Built and deployed fishing effort detection models achieving high accuracy in classifying vessel behavior from sensor streams, enabling near-real-time compliance monitoring across 10,000+ vessels in 18 countries. Spearheaded SecondScreen, a GIS web application for Fisheries Monitoring Centers visualizing vessel tracks, animal movements, and environmental data layers in real time; established product roadmap in close collaboration with marine scientists and fisheries management agencies. Managed full hardware/software lifecycle for NetSenz and GearSenz marine sensors – from field experiment design and validation protocols to IEEE-compliant software requirements specification. Conducted trajectory analysis in QGIS to identify critical marine habitats and candidate marine protected areas. Earlier role (Jun–Dec 2020) delivered interactive geospatial analytics dashboards (Streamlit, Dash, PowerBI) for vessel captains and fisheries enforcement officers.

Spotify – Stockholm, Sweden

Machine Learning Engineer – Internship Jul 2021 – Sep 2021

Built a hybrid playlist recommendation engine combining matrix factorization (SVD), neural collaborative filtering (NCF), and gradient boosting (XGBoost) for user engagement prediction. Applied TF-IDF and cosine similarity for content-based filtering on playlist metadata, achieving 15% improvement in recommendation accuracy over baseline.

Greenlytics – Stockholm, Sweden

Machine Learning Engineer – Internship Jan 2020 – May 2020

Developed a short-term solar irradiance forecasting model integrating Numerical Weather Prediction (NWP) forecasts, clear-sky irradiance models, and METEOSAT/SEVIRI satellite imagery. Applied deep learning and computer vision techniques including the SICSS algorithm to extract irradiance values under varying solar elevations and atmospheric conditions. Trained deep neural network models that outperformed ECMWF baseline predictions in accuracy – directly relevant to climate modeling at the ocean–land–energy nexus.

Primark d.o.o – Zagreb, Croatia

https://primark.hr/automatizacija/

Machine Learning Engineer – Industrial Automation Sep 2018 – May 2020

Developed CNN-based computer vision systems for manufacturing defect detection (misalignment, soldering errors, missing components) using TensorFlow, PyTorch, and OpenCV. Gained foundational experience in production ML pipelines and neural network optimization.

CONFERENCE PRESENTATIONS, POSTERS & TALKS

UN One Ocean Science Congress – Poster Jun 2025

Nice, France

"Using Machine Learning to Classify Fish Species in order to Scale Landing Declaration Systems." Presented computer vision approaches for automated fish species identification to improve fisheries data compliance at scale. Demonstrated how deep learning-based classification can integrate with digital Landing Declaration

Systems (LDS) to enable accurate, low-friction catch reporting – reducing data gaps in global fisheries stock assessments and supporting sustainable resource management. bluetraker.com/landing-declaration-system

Invited Talk Apr 2025

Stanford GSB – Taking the Pulse of the Planet: Sensors to Solutions

"The Missing Piece: How Fishing Fleets Can Complete Our Ocean Observing System." Addressed an interdisciplinary audience of researchers, policymakers, and industry leaders on leveraging commercial fishing vessels as platforms for scalable ocean data collection. Presented novel approaches to integrating fishing fleet operations – including vessel-mounted environmental sensors and real-time telemetry – with global ocean monitoring networks such as Argo and GOOS, to enhance climate and environmental surveillance in undersampled ocean regions. gsb.stanford.edu/events/taking-pulse-planet-sensors-solutions

Fireside Chat Oct 2024

Mountain View, California

"How Non-Terrestrial Networks (NTN) Will Impact the IoT Market and Adaptation Strategies for Integrators." Discussed satellite and NTN connectivity strategies for maritime IoT, drawing on experience deploying ocean-based sensor networks.

GZS Innovation Day – Keynote Speaker Sep 2024

Chamber of Commerce and Industry of Slovenia

Keynote following BlueTraker's national innovation award for marine sensor systems. media.gzs.si – GZS Innovation Award

Publications

[1] Peter Mastnak, Zlatko Mastnak, Iztok Pirnat, Uros Kac, and Elis Purkov. Innovating fisheries management and ocean observation: BlueSenz sensors turn fishing vessels into ocean data platforms. In *One Ocean Science Congress 2025*, Nice, France, 2025. doi:10.5194/oos2025-991.

[2] Iztok Pirnat, Zlatko Mastnak, Peter Mastnak, Nenad Hercigonja, and Uroš Kač. Challenges and solutions in quantifying fishing effort for passive gear. In *One Ocean Science Congress 2025*, Nice, France, 2025. doi:10.5194/oos2025-1121.

[3] Nik Vaessen, Peter Mastnak, and Sri Datta Budaraju. [re] latent weights do not exist: Rethinking binarized neural network optimization, 2020. Submitted to NeurIPS 2019 Reproducibility Challenge.